

UNIFIED MECHANICS

A WORLD OF DISTINCTIONS

Compact Monograph

One physical framework. One mathematical reconstruction. One explicit bridge.

FINAL TWO-PAPER EDITION | COMPACT MONOGRAPH

This edition consolidates the physical framework, the mathematical reconstruction, the observational language, the multiverse constraint algebra, and the empirical programme into one dependency-ordered account. The physical framework is the source semantics. The mathematics formalises, compresses, and tests it.

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STATUS OF THIS PAPER

This document introduces no independent theory. Every definition, equation, status grade, and empirical claim is a compression of the Final Main Monograph. Where detail is omitted, the Main Monograph governs.

The framework in one paragraph

A realised system remains identifiable because some distinctions are retained while local conditions select admissible changes. Those changes affect boundaries, interactions, and records. Unified Mechanics began as that physical picture: propagation carries state, accumulation retains structure, and the interface between them constrains both. The mathematics compresses the same operation into whole, part, remainder, a golden self-similar closure, two complementary rates, a three-path partition, and Primordial Loop Algebra. The physical realisation maps the three paths to Light, Boundary, and Matter; boundary response to gravity; saturated anchors to black holes; and adjacent loop mismatch to the dark boundary shear visible in galaxies and cosmology.

READING RULE

Observation motivated the source model. Mathematics formalises it. Physical conclusions require an explicit carrier, coupling, local condition, record, and inference. A numerical match alone is not a mechanism.

1. The physical source model

1.1 Realised systems

The empty description does not create a physical universe. A realised system requires a nonempty carrier and at least one distinction that persists long enough to constrain a later state.

$$\text{system} = \text{retained identity} + \text{parts} + \text{boundary} + \text{local condition} + \text{process} + \text{record} \quad (1)$$

Role	Question
Retained identity	What remains the same?
Parts	What relations realise the whole?
Boundary	Where is a difference or transfer resolved?
Local condition	What selects the actual behaviour here?
Process	What admissible change occurs?
Record	What persistent physical difference remains?

H₂O illustrates the rule. Molecular identity does not determine phase or trajectory by itself. Temperature, pressure, fields, neighbours, history, and boundary conditions select whether the realised water system rotates, vibrates, reacts, freezes, boils, or remains bound. A complete physical claim therefore includes both identity and local condition.

$$\text{mechanism} \rightarrow \text{coupling} \rightarrow \text{physical change} \rightarrow \text{record} \rightarrow \text{inference} \quad (2)$$

1.2 Self-description and Universopedia

A system describes itself operationally when what it retains and the conditions it occupies alter its boundary or records in a reconstructable way. Universopedia keeps that observation-first structure visible. The Familiarity Kernel preserves structured records that have not yet survived classification, reconstruction, and transfer.

NO SILENT PROMOTION

An internal normal form is not a physical measurement. A physical observation is not the proof of an internal theorem. They meet only through the realisation map.

1.3 The observed three modes

Mode	Physical action	Formal channel
Propagation	Carries state and geometric information outward.	W _L
Interface	Resolves exchange, inside/outside, tension, and record.	W _B

Mode	Physical action	Formal channel
Accumulation	Retains, binds, clusters, and builds persistent structure.	W_M

The interface is not seeded as a separate substance. It appears whenever propagation and accumulation coexist. The structure simulations implement this source model on several substrates; they test implementability and emergence, not the truth of the entire physical theory.

1.4 Anchors and black holes

Gravity is the boundary acting as a bidirectional constraint-anchor. Strategy A is stable organisation around an anchor. Strategy B is resolution into denser boundary infrastructure when the prior organisation cannot retain itself. Distributed halos and saturated black holes are different regimes of the same Strategy B object class.

2. The mathematical compression

2.1 Whole, part, remainder

$$(x+1)/x = x \Rightarrow x^2 = x+1 \Rightarrow \varphi = (1+\sqrt{5})/2 \quad (3)$$

The golden ratio is the unique positive solution of the minimal self-similar whole-part-remainder closure. The reciprocal relation is resolved across two boundary orientations:

$$r = 1/(2\varphi) = 0.309016994... \quad (4)$$

2.2 Three paths from two rates

$$1 = [r + (1-r)]^2 = r^2 + 2r(1-r) + (1-r)^2 \quad (5)$$

Channel	Weight	Operation
Matter	$W_M = r^2 = 0.0954915$	retained-retained / accumulation
Boundary	$W_B = 2r(1-r) = 0.4270510$	mixed / interface
Light	$W_L = (1-r)^2 = 0.4774575$	transmitted-transmitted / propagation

$$W_L + W_B + W_M = 1, \quad W_L/W_M = 5 \quad (6)$$

2.3 Dynamics and floor

$$r_{k+1} = 1/(4r_k + 2), \quad \mu = -4r^2 = -1/\varphi^2 \quad (7)$$

The forward map is attracting and alternating; the inverse is unstable. Under the physical clock realisation, that asymmetry supplies temporal orientation. A complete Light-Boundary-Matter record traversal retains the residual:

$$\varepsilon = r^3 = 0.029508497... \quad (8)$$

FLOOR DISCIPLINE

The floor applies only when the relevant physical traversal and record chain are identified. It is not a general tolerance band for any discrepancy.

3. Primordial Loop Algebra

A scale-free whole-part state is a positive projective pair (u,v) . The two primitive physical edits are accumulation and role reversal:

$$A:(u,v) \mapsto (u+v,v), \quad J:(u,v) \mapsto (v,u) \quad (9)$$

$$A:x \mapsto x+1, \quad J:x \mapsto 1/x \quad (10)$$

The Euclidean algorithm proves that A and J generate every admitted nonnegative unimodular projective lattice edit. With inverse translation admitted, the natural full setting is $\text{PGL}(2, \mathbb{Z})$. General periodic words generate periodic continued fractions and quadratic irrationals.

PRIMITIVE UNIVERSE RULE

A primitive closed loop has no proper inversion-delimited internal loop. It therefore contains exactly one inversion. Multiple-inversion words are composite internal braids.

$$M_n = A^n J = [[n, 1], [1, 0]] \quad (11)$$

$$\varphi_n = (n + \sqrt{n^2 + 4})/2 \quad (12)$$

Stage	Meaning
Π - bare accumulator	Unclosed substrate; no finite fixed point and no channel weights.
$A^{-1}J$ threshold	Repelling self-contact separatrix.
$n=0$ mirror	First closed but neutral unity state.
$n=1$ gold	First attracting primitive universe - ours.
$n=2$ silver	Next attracting child and adjacent upper boundary.

$$f'_n(\varphi_n) = -1/\varphi_n^2 \quad (13)$$

This corrected sequence separates the old overlapping uses of primordial, $n=-1$, $n=0$, accumulator, parent, and universe. Silver is our child and adjacent neighbour, not our parent.

4. The representation bridge

$$(\text{pure propagation, mixed interface, pure retention}) \mapsto (\text{Light, Boundary, Matter}) \quad (14)$$

The channel names are justified by what the formal paths do. Physical use then requires a carrier and field dynamics. A minimal metric realisation weights curvature by a local response field λ :

$$S = \int \sqrt{-g} [\lambda R / (16\pi G) - Z(\lambda)(\partial\lambda)^2/2 - U(\lambda) + L_m] d^4x \quad (15)$$

$$\lambda G_{\{\mu\nu\}} + (g_{\{\mu\nu\}} \square - \nabla_\mu \nabla_\nu) \lambda = 8\pi G (T_{\{\mu\nu\}} + T^{\lambda}_{\{\mu\nu\}}) \quad (16)$$

Gravity is the bidirectional boundary response carried by this metric equation. Where λ varies, its derivatives are part of the force and lensing mechanism; they cannot be dropped while retaining only a local rescaling of G .

5. The dark mechanism

Description	Place in one chain
Adjacent recursion	Supplies a persistent external boundary condition.
Boundary field	Carries that condition into our metric.
Strategy B anchor	Retains the condition as local or distributed infrastructure.
Hidden curvature	Describes the metric effect not reported as visible matter.
Apparent dark mass	Is the Newtonian reconstruction of that metric effect.
Second E_8	Is a proposed microscopic encoding of the orthogonal dark sector.

ONE MECHANISM

These phrases are different resolutions of the same causal chain. The global dark density budget and the local galaxy radial law are distinct outputs of that chain, not rival explanations.

6. Silver boundary shear and galaxies

Gold and silver reverse orientation at different contraction rates and carry different boundary fractions. Their exact mismatch is:

$$\Delta B_{12} = W_B^{(1)} - W_B^{(2)} = 0.0986238584... \quad (17)$$

Permanent adjacent contact turns that mismatch into boundary shear - "rubbing" in plain language. Silver multiplicative shells $R_{k+1} = \phi_2 R_k$ make equal work per shell accumulate logarithmically:

$$\Phi_I \propto \ln(R/R_0), \quad g_I \propto 1/R \quad (18)$$

With the Hubble recurrence as dimensional clock, the minimal acceleration is:

$$a_{12} = cH_0 \Delta B_{12} = 6.908558055369e-11 \text{ m s}^{-2} \quad [H_0 = 72.1 \text{ km s}^{-1} \text{ Mpc}^{-1}]$$

The minimal bilinear boundary closure uses the baryonic field g_b , total field g , and interface remainder $g - g_b$:

$$g(g - g_b) = a_{12} g_b \quad (20)$$

$$g = [g_b + \sqrt{(g_b)^2 + 4a_{12}g_b}] / 2 \quad (21)$$

$$g_b \ll a_{12}: \quad g \approx \sqrt{(a_{12}g_b)}, \quad v^4 = GM_b a_{12} \quad (22)$$

$$M_{\text{app}}(R) \propto R, \quad \rho_{\text{app}}(R) \propto 1/R^2 \quad (23)$$

STATUS

The PLA mismatch and shell geometry are derived internally. Their physical shear interpretation, cosmological clock coupling, and bilinear constitutive law are explicit conditional/proposed realisations. They are fixed enough to be falsified by joint rotation, lensing, cluster, morphology, and full-SPARC tests.

7. Cosmological closure

$$\Omega_b = r^2/2 \quad (24)$$

$$\Omega_{DM} = 4r^2(1-r) \quad (25)$$

$$\Omega_{DE} = 1 - 9r^2/2 + 4r^3 \quad (26)$$

$$\Omega_b + \Omega_{DM} + \Omega_{DE} = 1 \quad (27)$$

The density budget states how channel content is globally apportioned. The silver law states how boundary response is radially distributed around a recorded baryonic source. The two calculations answer different questions and are now kept separate.

The cosmological-constant notation is also separated: r^{240} is the internal E_8 traversal scale; r^{241} is the recorded residual after one final boundary registration. The physical Hubble split uses the later transmission-corrected expression $W_M(2/\sqrt{5})$, while $3r^3$ remains the historical first-order estimate.

8. Quantum, particles, and living systems

Quantum registration is a boundary process: a propagated state is coupled, resolved, and retained as a record. The golden eigenmode identity gives a per-cycle survival coefficient $1/\varphi$, but its physical use requires an isolated measurement protocol. Quantisation is treated as completed traversal counting; the dimensional map to $\hbar$ remains a named derivation.

$$1 - 1/\varphi^2 = 1/\varphi \quad (28)$$

Particle formulas are retained in the Main Monograph's graded ledger. Their arithmetic does not by itself establish the physical mode assignment. The $E_8 \times E_8/G_2$ construction is a proposed microscopic realisation, not the axiom from which the systems framework depends.

Living systems are retained identities that actively regulate their boundaries and use records to change later processes. Language is a higher-order version of the same operation: distinctions propagate, relations are resolved, and another system retains a reconstructable record.

9. Tests and the old galaxy null

THE OLD SPARC RESULT

UM-Q and a generic quadratic produced identical results because they were the same basis with different labels. That result is the quadratic compiler baseline equivalence. It did not test the silver boundary operator and is not a null result for the physical galaxy mechanism.

Pilot model	Galaxy-balanced acceleration RMSE
Baryons only	0.527467 dex
Fixed RAR	0.240355 dex
Previous UM age/light	0.237422 dex
Silver-interface closure	0.223741 dex

The silver law improved the 12-galaxy, 270-point pilot and beat fixed RAR in 8 of 12 galaxies, but the paired bootstrap interval crossed zero. The correct conclusion is that the zero-fit physical operator survived a small pilot and now requires a frozen full-catalogue and joint-lensing execution.

10. Final closure

THE COMPLETE CHAIN

realised distinction → retained identity → local condition → propagation/accumulation/interface → whole-part-remainder closure → $\varphi \rightarrow r \rightarrow$ three paths → PLA constraint lattice → explicit physical realisation → boundary gravity → adjacent-rung shear → records and tests

The final two-paper corpus no longer asks the reader to guess whether the physics came from the mathematics or the mathematics from the physics. The physical framework is the source model and historical discovery. The mathematics is its formal compression and internal constraint. The realisation bridge states exactly where physical assumptions enter. Every claimed observable is attached to a carrier, mechanism, coupling, local condition, record, and test.

Grade	Examples in this compact paper
Proved	Golden positive closure, partition identity, positive PLA completeness, primitive metallic spine, stability multipliers.
Conditional	r normalisation, channel realisation, traversal floor, scalar-tensor boundary interpretation, cosmological accounting.
Proposed	Silver bilinear galaxy law, registration experiments, microscopic particle encodings.
Conjectured	Specific $E_8 \times E_8 / G_2$ microscopic completion and broader cross-scale identifications.
Postulate	Physical realisation, local self-description, carrier/record requirement.

GOVERNING DOCUMENT

The Final Main Monograph contains the proofs, mechanisms, formula ledger, provenance map, and falsification surfaces. This Compact Monograph is a strict compression and introduces no separate branch of Unified Mechanics.